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IIIrd-Semester B.Tech Course
Mid Semester Examination, October-2021

COURSE CODE: ICECC08/ECE IOT/ECE/EE

Course Title: DIGITAL CIRCUITS AND SYSTEMS

Time: 1 hour 30 Minutes

Maximum Marks: 25

Note: Attempt all questions. Each question carries equal marks.

Q.No.	Questions	Marks	CO
1a.	1101010 is a signed binary number. Determine the decimal value in each case, if it is in (i) sign-magnitude form, (ii) 2's complement form, and (iii) 1's complement form.	2.5	01
1b.	Design a logic diagram of a full subtractor using only 2 input NOR gates.	2.5	01
2a	What is race around condition in JK flip-flop and how do you eliminate it?	2.5	02
2b.	Realize T-FF from JK-FF.	2.5	02
3a.	Minimize the following boolean function using K-map: $F(A, B, C, D) = \sum m(1, 3, 4, 6, 8, 9, 11, 13, 15) + \sum d(0, 2, 14)$	2.5	01
3b.	Obtain 3 level NOR – NOR implementation of $f(a,b,c,d,e,f)=[ab+cd]ef$.	2.5	01
4a.	Implement 4 to 16 decoder using 3 to 8 decoders.	2.5	02
4b.	Convert the given expressions in the canonical form. $Y(X, Y, Z) = X \cdot (X + Y) \cdot (X + Y + Z)$	2.5	01
5a.	Explain how a master slave JK Flip Flop works?	2.5	02
5b.	Simplify the following Boolean function by using a Quine McCluskey method. $F(A, B, C, D) = \sum m(0,2,3,6,7,8,10,12,13)$	2.5	01